

# RESSCOTT –INSPECTION REPORT

JOB #: IR-RES0323-41

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## SUBJECT:

**Checklist on Solar System:  
Inspection conducted on Solar System Components for  
60KW Hybrid Solar PV System**

CERTIFICATE#: IR-RES0323-41

DATE: 23/07/24

|                                   |                                |
|-----------------------------------|--------------------------------|
| OCCUPIER                          | Regan Pirali                   |
| ADDRESS                           | 44 Mayfeild, Valsayn, Trinidad |
| DESCRIPTION OF SOLAR SYSTEM       | 60 KW Hybrid Solar PV System   |
| LOCATION                          | SAME AS occupier               |
| SERIAL # / LOCAL #                | --                             |
| RATED CAPACITY                    | --                             |
| DATE OF CONSTRUCTION              | 23/07/24                       |
| DATE OF LAST EXAMINATION          | NEW                            |
| HOW LONG EQUIPMENT WAS IN SERVICE | Newly installed                |

## TYPE OF OPERATION:

|                        |                                                                                                                                                    |
|------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| How often system works | Daily <input checked="" type="checkbox"/> Weekly <input type="checkbox"/> Monthly <input type="checkbox"/> Intermittently <input type="checkbox"/> |
| Last Repaired          | By OEM <input type="checkbox"/> Locally <input type="checkbox"/> No Previous Record <input type="checkbox"/>                                       |

## TYPE OF FACILITY:

|                         |                          |                                     |
|-------------------------|--------------------------|-------------------------------------|
| Car park                | Home / Residential       | <input checked="" type="checkbox"/> |
| Warehouse (internal)    | Industrial Estate        |                                     |
| Commercial              | Park                     |                                     |
| Laydown yard            | Sporting ground          |                                     |
| Roadway /walk ways      | Stadium                  |                                     |
| General social compound | Other – (Container Unit) |                                     |



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## INVERTER AND BATTERY INSPECTION:

| Component                                   | N/A | Tested And Ok | Faulty | Comments                                                                       |
|---------------------------------------------|-----|---------------|--------|--------------------------------------------------------------------------------|
| Controls                                    |     | ■             |        |                                                                                |
| Push Buttons                                |     | ■             |        |                                                                                |
| Fan/cooler                                  |     | ■             |        |                                                                                |
| Operation                                   |     | ■             |        | Continuous                                                                     |
| Positive/ Negative                          |     | ■             |        | Intact and secured                                                             |
| Upper Limit trip                            |     | ■             |        | Breakers tested ok                                                             |
| Lower Limit trip                            | ■   |               |        |                                                                                |
| PV input connection                         |     | ■             |        | In accordance to PV load                                                       |
| Load output                                 |     |               |        | Intact and secured. 70 mm <sup>2</sup> 4 core cable used @ 120/240V @ 200 amps |
| Safety devices                              | ■   |               |        |                                                                                |
| Utility charge current for hybrid operation | ■   |               |        | 240 V supply for charge current. In accordance to manufacture charge rate - NA |
| Battery bank wiring                         | ■   |               |        | System is grid Tied with ZERO-GRID feedback @ self-consumption mode.           |
| Battery bank covering                       | ■   |               |        |                                                                                |

## SOLAR PANELS AND RACKING:

| Maintenance/Condition/Operation                                                                                              |                                                         |
|------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------|
| Are the following parts of the PV array properly maintained and in good order? If not state what defects have been observed. | In Good Order – No defects found.                       |
| (a) Enclosure – combiner box                                                                                                 | Clear and in good condition                             |
| (b) Piping / conduit run                                                                                                     | Intact and secure                                       |
| (c) PV array Racking                                                                                                         | Intact and secure                                       |
| (d) PV panels                                                                                                                | In good condition and clean                             |
| (e) PV wiring to combiner box.                                                                                               | Safe and secure.                                        |
| (f) Over running devices                                                                                                     | Functional                                              |
| (g) Other electrical equipment                                                                                               | Functions and power rating were found to be acceptable. |
| (h) PV Frame (ground mounted or roof mounted)                                                                                | Roof – Unistrut galvanized framing.                     |
| (i) Other parts                                                                                                              | ----                                                    |

What parts were inaccessible? (if any): None

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|                                                                                               |
|-----------------------------------------------------------------------------------------------|
| Comments: An inspection and service must be conducted every six (6) months to ensure warranty |
|                                                                                               |
|                                                                                               |

**DURING INSPECTION/REPAIR:**

What issues were observed (if any)?

|      |
|------|
| NONE |
|------|

Notice of Defects during Repair / Service (if any):

|      |
|------|
| NONE |
|      |

**REPAIRS / SERVICE DONE:**

Description of parts repaired (List part #):

NONE – NEW system installed

**PHOTOS:**

REFER TO Attached

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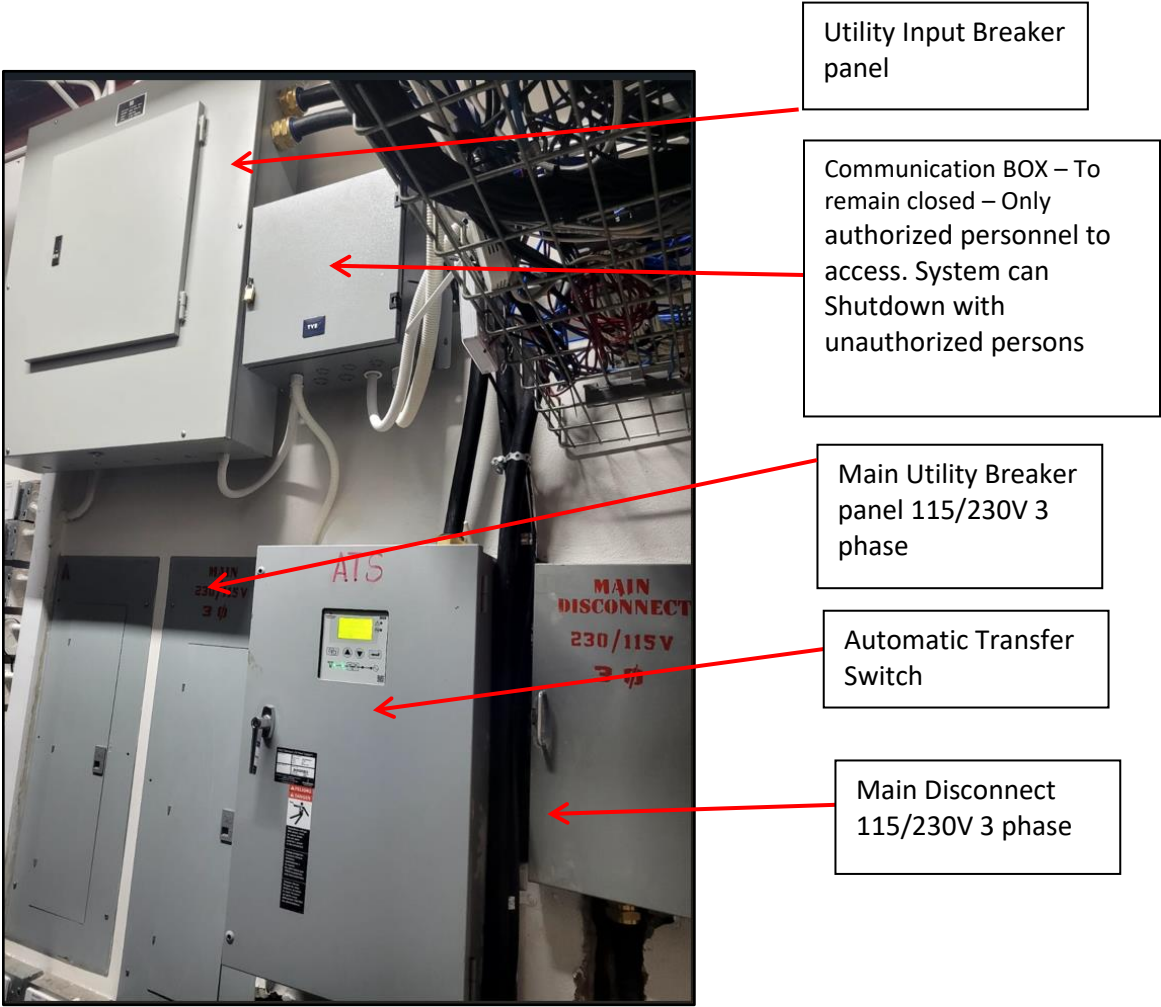


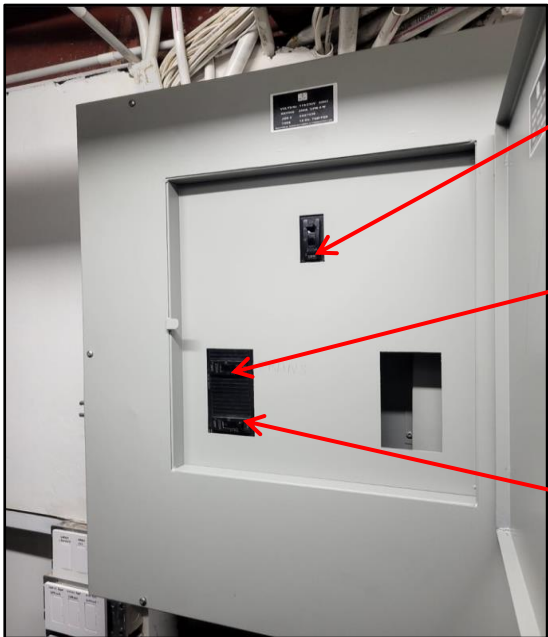
Photo #1: Overview of panel layout in communications room.



Main Disconnect, 200 Amp breaker from Utility connection. This breaker shut down the house or in case power goes off, the Back-up generator is initiated.

The back up generator does not affect the solar PV equipment.

Photo #2: Overview of Main disconnect Breaker



Main breaker 200 amp, Incoming from Main Utility breaker to split into distribution for transformer and loads

Isolator from Solar inverter Transformer

Isolator to main load distribution panel

Photo #3: Overview of Utility Splitter Breaker panel

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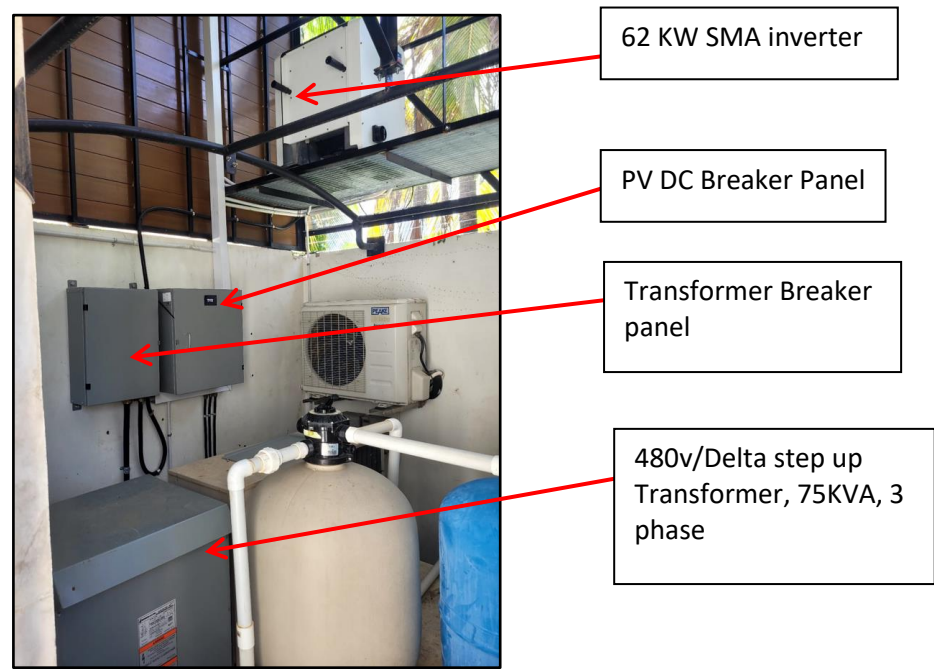


Photo #4: Overview of solar components

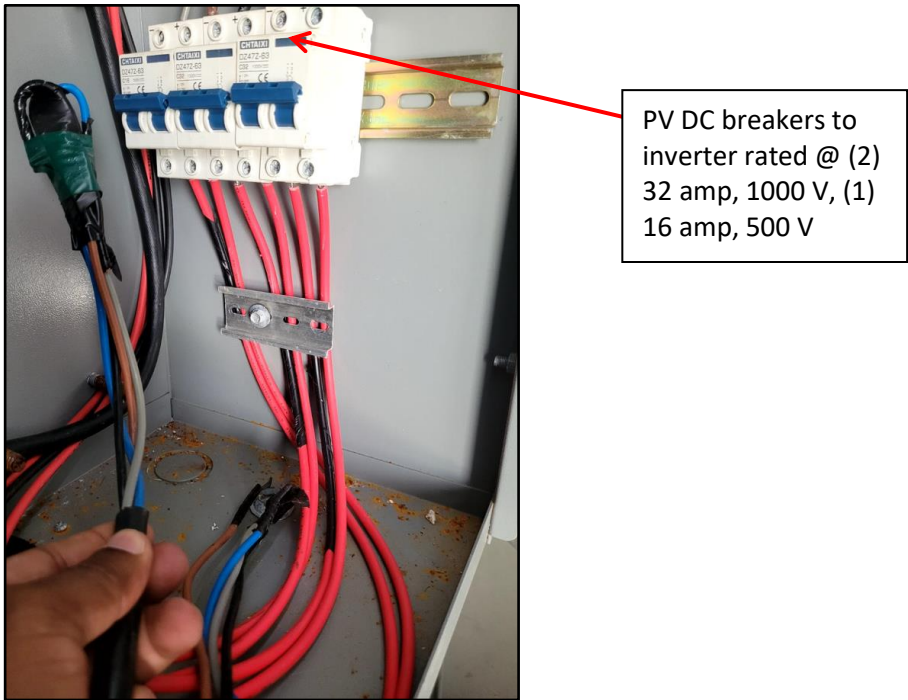


Photo #5: Overview of PV DC breakers

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Shut down Isolator  
for transformer to  
inverter, switch can  
be used to shut down  
solar system

**Photo #6: Overview of shut down isolator, transformer to inverter**

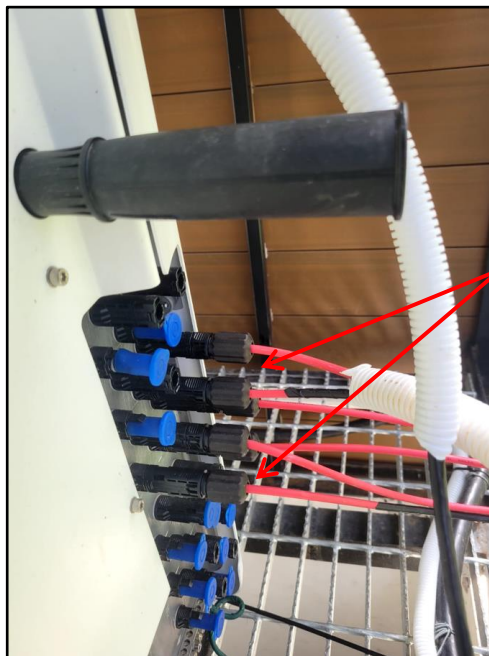


62 KW SMA  
inverter

**Photo #7: Overview of SMA Inverter**

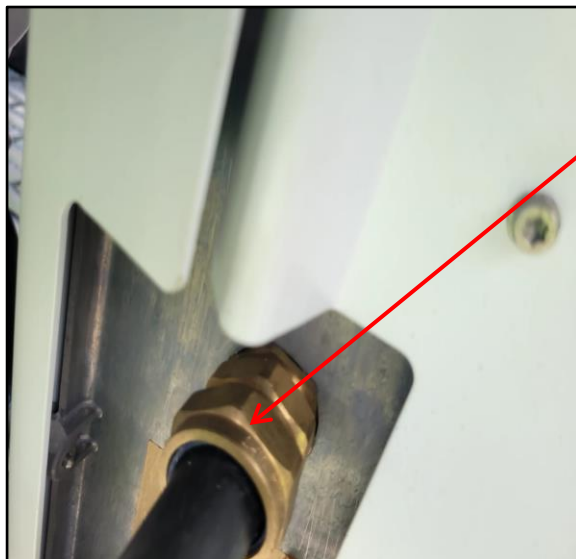
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PV connection to  
SMA inverter, 3  
arrays, 2 arrays of 13  
panels and 1 array of  
6 panels

**Photo #8: Overview of PV connection to inverter**

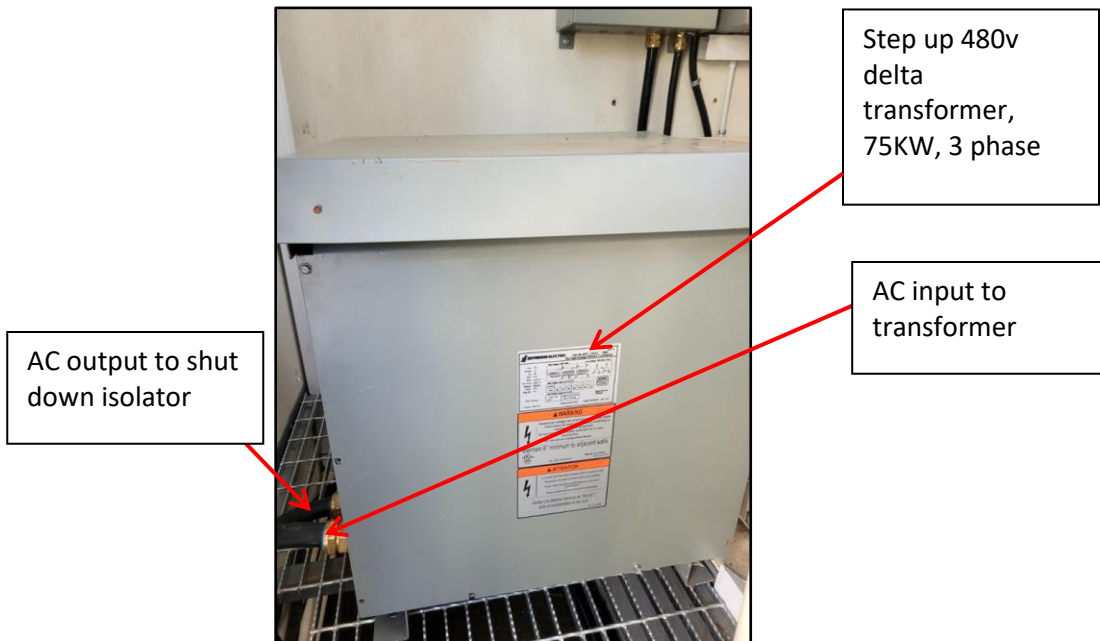


AC input to  
Inverter @ 480 V  
from step up  
transformer

**Photo #9: Overview of AC connection to inverter**

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**Photo #10: Overview of Step-up Transformer**



**Photo 11 A**

PV arrays, 570 W solar panels.

32 panels in total, consisting of:

- a) 2 strings @ 13 panels in one (1) String
- b) 1 string @ 6 PV panels. (see gazebo)



**Photo 11 B**

PV arrays, 570 W solar panels at Gazebo  
1 string @ 6 PV panels.

**Photo #11, A and B: Overview of PV array.**

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## **THE BENEFITS TO THIS SERVICE:**

- Repairs are made to manufacturer specifications or change out is done
- Maintains the equipment integrity
- Complies with Manufacturer/OSHA regulations
- Maintains Warranty
- All work done will be states on a service certificate and used for history.

## **LIMITATIONS:**

- None

## **RECOMMENDATIONS:**

- Client to conduct 6 month assessment on to ensure safe operation and to identify if any abnormal operation are present. More frequent cleaning will be dependent on the client view of cleaning and condition at the location.

## **NEXT SERVICE EXAMINATION DUE: 20/01/25**

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## DISCLAIMER:

\*\*\*\*\*Declaration\*\*\*\*\*

I \_\_\_\_\_ accepted the product of RESSCOTT Limited in good working condition.

All components and fire safe elements were identified to me and I agree on the terms and conditions of this product.

\*\*\*\*\*Terms and Conditions\*\*\*\*\*

The RESSCOTT company contains detailed inspection photos and reports that is kept in-house to identify any tamper or alterations that may be observed. The said report will be used as a comparison during inspection.

The Client/user must operate the product within its specifications and standard operating Procedure (SOP).

The product is not to be used outside its specifications or SOP to hinder the safety and compromise any fire safe elements or warranty.

All components of this product are designed to specific needs of the client and shall not be altered in any form or fashion. Alterations done by the client can lead to failure and immediately void the warranty and trade name of the supplier.

Clients are advised that they must review their documentation before acceptance of our products/ services, as RESSCOTT Limited is not responsible for any discrepancies thereafter.

Mobilization and technicians fee must be charged for when items on warranty are to be changed out.

RESSCOTT Ltd can provide virtual diagnostic services to ensure the client satisfaction. Any defects noted, the equipment can be changed out in accordance to its warranty period.

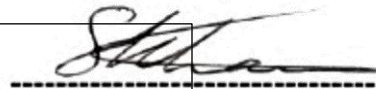


**GRAEME JONES**

**MANAGING DIRECTOR**

MBA PROJECT MANAGEMENT  
BSC. MECHANICAL ENGINEERING (UWI)  
REGISTERED PROFESSIONAL ENGINEER (BOETT)  
QA/QC INSPECTOR  
ASQ – CRE, CMQ / OE, CSSGB  
PMI – PMP  
ASNT LEVEL 1  
FAILURE ANALYSIS ENGINEER

Signed signatures. Call for verification



**SHAMIIR KHAN**

**OPERATIONS DIRECTOR**

BSC. MECHANICAL ENGINEERING (UWI)  
QA/QC INSPECTOR  
CERTIFIED - CWI INSPECTOR, NACE, ASNT LEVEL II, API 510  
INSPECTOR  
METALLOGRAPHIC INTERPRETATION (ASM  
INTERNATIONAL)